

The Vel Protocol: Optimization of Decentralized Swarm Coordination for Autonomous Area Coverage

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Abstract

This letter addresses the challenge of decentralized multi-robot area coverage in unknown environments, where agents must self-organize without centralized control. We present the **Vel protocol**, a decentralized coordination system that introduces *Map-Aware Gradient Repulsion* (MAGR). By steering agents toward the nearest locally unexplored grid cells via a binary shared occupancy map, MAGR replaces traditional center-outward bloom strategies which typically suffer from boundary-induced congestion and poor population scaling.

Through five systematic parameter sweeps across 105,000 simulation trials, we characterize the interactions between agent population (N), repulsion factor (G), momentum coefficient (μ), and MAGR scan radius (r). Our findings reveal a population-dependent optimum where $G^* = 1.5$ for $N = 300$ and $G^* = 2.75$ for $N = 150$. Critically, we identify the momentum coefficient as the dominant efficiency driver; $\mu = 0.99$ yields a 49.6% efficiency gain over baseline assumptions while minimizing variance ($\sigma = 0.089$). Further, we establish that MAGR is radius-invariant for $r \geq 2$, identifying $r = 2$ as the computational optimum. The global system optimum achieves a 78.7% improvement over baseline performance. Closed-form expressions for G^* , μ^* , and r^* are derived and validated against empirical results.